

Makda T. Mammo

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🎓 EDUCATION

M.S. in Science and Technology Policy (STP)

KAIST (Korea Advanced Institute of Science and Technology) [🔗](#)

Daejeon, South Korea

08/2026 – Present

- Focus areas: AI governance, technology policy, and the societal impacts of AI

B.Sc. in Software Engineering CGPA 3.97/4.0

Mekelle University [🔗](#)

Mekelle, Ethiopia

09/2019 – 07/2026

Relevant coursework

- Data structures and algorithms, Artificial intelligence, machine learning, cloud computing, compiler design, project management, web design and programming

Kifiya AI Mastery (KAIM) Training Program

10 Academy, Kifiya & Mastercard Foundation [🔗](#)

Remote

11/2025 – 02/2026

- Selected for a competitive, fully-funded 12-week remote program (up to 40 hrs/week) in partnership with Mastercard Foundation
- Specialized in Generative AI Engineering, Machine Learning Engineering, Data & Analytics Engineering, MLOps & AutoML, Prompt Engineering

👛 PROFESSIONAL EXPERIENCE

AddisCoder [🔗](#)

Co-Head Teaching Assistant

Addis Ababa, Ethiopia

07/2026 – Present

- Coordinated teaching assistants and supported the delivery of an intensive computer science program.
- Led lab logistics, TA coordination, and student support across multiple computer labs.
- Supported curriculum preparation, debugging, and instructional material development.

Teaching Assistant

07/2025 – 08/2025

- Enhanced student understanding of Data Structures and Algorithms by leading lab sessions for 23 students.
- Improved learning outcomes by creating and refining 4 Jupyter Notebooks daily, offering tailored exercises.
- Professors: Jelani Nelson (UC Berkeley/Google), Daniel Kang (UIUC), Huy Nguyen (Northeastern), Huacheng Yu (Princeton)

Mozilla Festival - GirlsWhoML Sprintship

Research & Technology Intern

Remote

11/2025 – 12/2025

- Selected for a competitive research sprintship examining the societal impacts of machine learning, with a focus on bias, fairness, and disparate effects on women and marginalized communities.
- Conducted collaborative research and primary data collection on algorithmic fairness, dataset representation, and real-world model evaluations.
- Contributed to frameworks and resources for translating technical ML challenges into actionable insights for more equitable and responsible technology policy.

Buna Labs

Backend Engineer

Addis Ababa, Ethiopia

01/2025 – 07/2026

- Designed and implemented backend systems enabling seamless integration with frontend applications
- Built secure authentication systems
- Collaborated with frontend developers to integrate Django backend with React/Vue.js
- Integrated Redis to cache highly accessed profile data, improving scalability and reducing API response time by over 20%.

- I developed a rigorous testing pipeline, authoring over 200 unit and integration tests using Pytest, achieving 85% test coverage
- Technologies used: Python, Django REST Framework, Celery, Redis, JWT, Render, gunicorn,

SKILLS

- Python, Java, C++, C#, Django, Django REST Framework, PyTorch, Flutter, Firebase, Supabase, Redis, Celery, AWS, Docker, Git, GitHub Actions, Machine Learning, Data Structures & Algorithms

PROJECTS

TigrinyaNLP

12/2025 – Present

- Built an open-source preprocessing framework for Tigrinya to enable reproducible NLP research in low-resource, morphologically rich settings.
- Formalized script-aware preprocessing for Ethiopic text, including Unicode normalization, punctuation boundary modeling, and whitespace canonicalization.
- Designed configurable cleaning and stopwords policies to support downstream tasks such as topic modeling and supervised learning, mitigating representation-level exclusion in an under-supported language.
- **Technologies:** Python, Unicode (NFC), Regex-based Text Processing, PyPI Packaging,

MentorMe

10/2025 – Present

- Designed and developed a mobile platform connecting tutors, students, and parents to reduce information gaps in informal education markets.
- Built a Flutter client with role-based authentication and distinct workflows for tutors, learners, and parents.
- Implemented a hybrid recommendation engine combining content-based filtering and matrix factorization to match users based on profiles and interaction history.
- Integrated real-time messaging and tutoring request tracking to model and manage tutoring relationships.
- **Technologies:** Flutter (Dart), Firebase Authentication, Cloud Firestore, Matrix Factorization, Content-Based Filtering, Real-Time Messaging

Breast Cancer Histology Image Classification

02/2025 – 04/2025

- Designed a two-stage PyTorch-based CNN pipeline that classifies breast cancer histology images into Normal, Benign, InSitu, and Invasive categories.
- Implemented patch-wise feature extraction followed by image-level aggregation to handle high-resolution medical images (2048x1536).
- Achieved reliable predictions using pretrained weights, test-time augmentation, and ROC/AUC analysis.
- Technologies: PyTorch, Computer Vision, Deep Learning, CNNs, Transfer Learning, Test-Time Augmentation

CERTIFICATES

GirlsWhoML Research and Technology internship Completion

AddisCoder TA

100 Days of Code: The Complete Python Pro Bootcamp

- Content: Python basics, OOP, Web development, Data Science and Machine Learning
- Instructor: Dr. Angela Yu